



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

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SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510; Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Product name: bispyribac-sodium 10% W/V SC

Product use: Herbicide

Relevant identified uses of the substance/mixture and uses advised against:

Agricultural use

2. HAZARD IDENTIFICATION

Emergency overview:

Harmful by inhalation. May cause sensitization by skin contact.

Adverse Human Health Effects

Effect of over exposure:

Over-exposure experiments on animals showed that the administration of an extra-ordinary large quantity of Bispyribac-sodium 10% W/V SC may cause hypoactivity, abnormal walking and prostration.

Environmental hazards:

Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

3. COMPOSITION/INFORMATION ON INGREDIENT

Substances: Formulation

Chemical name: 2,6-bis[(4,6-dimethoxy-2-pyrimidinyl) oxy] benzoic acid

CAS Number: 125401-92-5

Ingredients and composition:

Active ingredient $\geq 10\%$ W/V

- bispyribac-sodium

Inert ingredient $< 90\%$ W/V

- Surfactant



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- Water

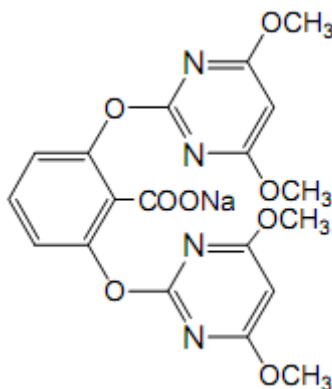
- Etc.

Chemical family: Pyrimidinyl carboxy compounds

Molecular formula: $C_{19}H_{18}N_4O_8$

Molecular weight: 452.4

Structural formula:



4. FIRST AID MEASURES

Description of first aid measures

Have the product container, label or Material Safety Data Sheet with you when going for treatment.

Tell the person contacted the complete product name, and the type and amount of exposure.

Describe any symptoms and follow the advice given.

In case of skin contact:

Wash material off the skin in flowing water or shower with soap. If irritation persists, consult a physician immediately.

In case of eyes contact:

Immediately flush with copious amount of water for at least 15 minutes. If irritation persists, consult a physician.

In case of Ingestion:

Induce vomiting by giving one or two glasses of water. Get a medical attention immediately.

If the victim is unconscious or having convulsions, never give fluids nor induce vomiting.

In case of Inhalation:

Remove the victim to fresh air and keep him warm and quiet. Get a medical attention immediately. If not breathing, clear the airway and give artificial respiration. If breathing is difficult, give oxygen.

Note to physician:

No specific antidote. Apply symptomatic therapy.



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5. FIRE FIGHTING MEASURES

Flash Point and flammability: Not available/applicable

Extinguishing Media: Water, dry chemical powder, foam or carbon dioxide

Special Fire Fighter Procedure

- Use water, dry chemical powder or carbon dioxide in extinguishing.
- For a fire in surrounding area, apply water to cool and protect surrounding equipment. Remove containers, if feasible, from the fire area to safe place immediately.
- Fire fighters should wear protective gears including a positive pressure self-contained breathing apparatus (SCBA). Evacuate non-essential personnel.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

- Evacuate non-essential personnel by roping the area of leak or spill. Evacuate people topwind of the spilled area. Remove all sources of ignition; No flares, smoking, or flames in the area immediately. Wear protective gears and never work downwind.
- For small spills, adsorb with inert materials (e.g. dry sand or earth), then place in a waste chemical container for disposal. Flush residual spill (area) with a plenty of water.
- Prevent spills from entering sewers, watercourses or low areas.

7. HANDLING AND STORAGE

Precautions for safe handling: KEEP OUT OF REACH OF CHILDREN

- When handling and spraying the product, wear duty working clothes, cap, gloves, glasses and mask to avoid contact with skin, eyes and inhalation of dust, fog and vapors.
- After work thoroughly wash hands, face and possible exposed areas of body and change clothing.

Precaution for safe storage:

- Keep containers tightly closed and store in a cool, dark and dry place.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

Control parameters

ACGIH TLV (1992-93): Not established

Engineering controls

- The use of local exhaust ventilations is recommended.
- Containers should be kept tightly closed.
- Make available emergency showers and eye washes in a work area.



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Personal Protective Equipment

Wear chemical cartridge respirator with an organic vapor cartridge, airline respirator, positive pressure self-contained breathing apparatus, safety goggles, gloves and boots as appropriate.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Appearance:	Viscous suspension concentrate liquid
Color:	Off-white
Odor:	Almost odorless
Specific gravity:	1.059 – 1.070 (20 °C)
Viscosity:	300 – 600 mPa
Suspensibility:	85%
Dispersibility:	Good
Compatibility:	Compatible to almost all pesticides
Heat and cold stability:	Stable
pH:	6.0 – 8.0

10. STABILITY AND REACTIVITY

Flash Point:	Not applicable
Auto ignition Temperature:	Not applicable
Explosion Limit:	Not applicable
Stability and Reactivity:	Stable under normal conditions
Hazardous decomposition products:	None expected.
Hazardous Polymerization:	Not known to occur
Incompatibility:	Compatible to almost all pesticides
Storage Stability:	Stable for 3 years if stored in a cool, dark and dry place.

11. TOXICOLOGICAL INFORMATION

Information on toxicological effects

Acute toxicity

LD50 (oral):	> 5,000 mg/kg (rat) (male & female)
LD50 (dermal):	> 4,000 mg/kg (rat) (male & female)
LC50 (4h) (inhalation):	> 4.69 mg/L



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Irritant properties

Eye irritation: Not irritant in rabbit

Skin irritation: Mild irritant in rabbit

Allergenic and sensitizing effects: Not sensitizing in Guinea pig

12. ECOLOGICAL INFORMATION

Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the mixture, bispyribac-sodium:

Fish – Acute toxicity:

Rainbow trout,

LC50 (96h): > 100 ppm

Bluegill sunfish,

LC50 (96h): > 100 ppm

Daphniamagna,

LC50 (48 h): > 100 mg/l

Avian – Acute oral toxicity:

Bobwhite quail,

LD50: > 2,250 mg/kg

LC50 (5 d): > 5,620 mg/kg diet

Mallard duck,

LC50 (5 d): > 5,620 mg/kg diet

Algae – Acute toxicity:

Selenastrum capricornutum,

EC50 (120 h): 3.4 mg/l

NOEC: 0.625 mg/l

Earthworm – Acute toxicity:

Eisenia foetida,

NOEL (14 d): > 1,000 mg/kg soil

Honey bee – Acute toxicity:

LD50 (oral): > 200 µg/bee

LC50 (contact): > 7,000 mg/l spray



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13. DISPOSAL CONSIDERATIONS

- Dispose of in accordance with local regulation.
- Do not dump into sewers or into any body of water.
- Product disposal: Pesticide spray mixture or rinsate that cannot be used or chemically reprocessed should be disposed of in a landfill approved for pesticides. Dispose of in compliance with all state and local laws and regulation.
- Container disposal: Triple rinse (or equivalent) and dispose of in an incinerator or landfill approved for pesticide containers.

14. TRANSPORT INFORMATION

Check the leakage of containers before loading. Make sure to load containers correctly to prevent them from falling down

Follow all regulations with regard to transporting chemicals in your country.

This product is not classified as dangerous good.

DOT hazard class: NA.

UN number: Not regulated.

15. REGULATORY INFORMATION

Follow all regulations in your country.

R-phrases:

R43—May cause sensitization by skin contact.

R52/53—Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

S-phrases:

S24—Avoid contact with skin.

S35—This material and its container must be disposed of in a safe way.

S37—Wear suitable gloves.

S57—Use appropriate containment to avoid environmental contamination.

WHO (formulation): III (Slightly hazardous)



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16. OTHER INFORMATION

This SDS summarizes our best knowledge of the health and safety hazard information of the product and how to safely handle and use the product in the workplace. Each user should read this SDS and consider the information in the context of the how the product will be handled and used in the workplace including in conjunction with other products.

If clarification or further information is needed to ensure that an appropriate risk assessment can be made the user should contact the company.

THEP WATANA CO., LTD.